

HiveForce Labs

# THREAT ADVISORY



## ATTACK REPORT

### Threat Actors Using Windows Error Reporting tool to Deploy Pupy RAT

Date of Publication

January 5, 2023

Admiralty Code

A1

TA Number

TA2023007

# Summary

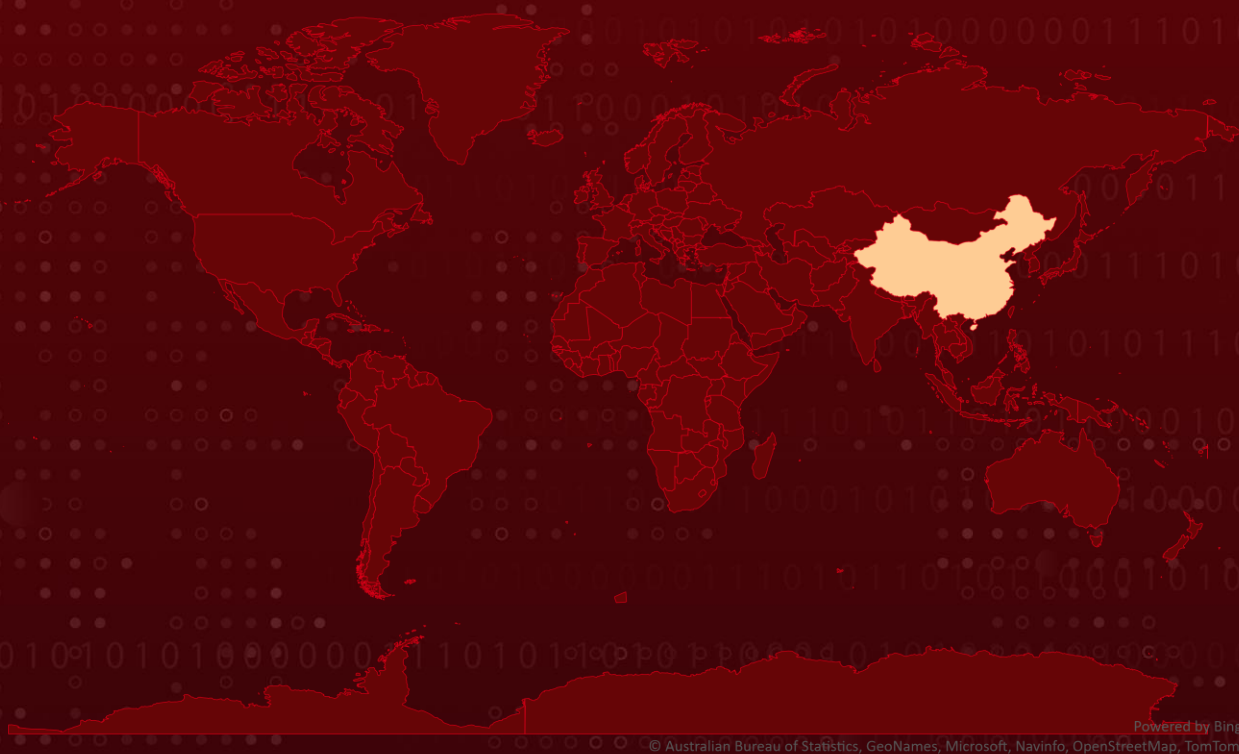
**First appeared:** 2013

**Attack Region:** China

**Impacted Platform:** Windows

**Attack:** The Pupy RAT malware is attempting to evade detection by disguising itself as the legitimate WerFault.exe process.

## Attack Regions



# Attack Details

## #1

The Pupy RAT malware is using a technique called DLL side-loading to disguise itself as the legitimate WerFault.exe process in order to evade detection. The malware is delivered via an ISO image that contains a malicious DLL file, a shortcut file, and an Excel file. When the shortcut file is opened, it runs the WerFault.exe process, which then uses the DLL side-loading technique to load and execute the malicious DLL.

## #2

This DLL contains a custom API resolving function and uses the CreateThread function to create two threads. The first thread opens the Excel file, while the second thread uses the SystemFunction032 function (which is undocumented in MSDN) to perform RC4 decryption on the data contained in the overlay of the DLL. The decrypted data is then loaded and executed in memory, resulting in the Pupy RAT being run on the victim's machine. It attempts to make a connection to a command and control server in the background while the victim believes that WerFault is running.

## #3

The Pupy RAT is an open-source, cross-platform remote administration tool that has been used by APT33 and APT35 since 2013, for cyber espionage purposes. However, it is not clear if this specific attack was carried out by these APT groups.

# Recommendations



### Security Leaders

Phishing simulations and routine education and awareness training and communications rarely account for MFA fatigue and web browser hygiene. Integrate and communicate all lessons learned.



### Security Engineers

- **Uni5 Users:** This is an actionable threat advisory in HivePro Uni5. Prioritize and block all indicators attributed to the threat actors and attacks through your Command Center. Test your controls with Uni5's Breach & Attack Simulation.
- **All Engineers:** Refer to and action upon the 'Potential MITRE ATT&CK TTPs' & 'Indicators of Compromise (IoC)' on the following pages.

## Potential MITRE ATT&CK TTPs

|                                                          |                                                  |                                                                |                                                        |
|----------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| <u><b>TA0002</b></u><br>Execution                        | <u><b>TA0011</b></u><br>Command and Control      | <u><b>TA0009</b></u><br>Discovery                              | <u><b>TA0006</b></u><br>Credential Access              |
| <u><b>TA0005</b></u><br>Defense Evasion                  | <u><b>TA0004</b></u><br>Privilege Escalation     | <u><b>TA0040</b></u><br>Impact                                 | <u><b>T1027</b></u><br>Obfuscated Files or Information |
| <u><b>T1059</b></u><br>Command and Scripting Interpreter | <u><b>T1574.002</b></u><br>DLL Side-Loading      | <u><b>T1574</b></u><br>Hijack Execution Flow                   | <u><b>T1036</b></u><br>Masquerading                    |
| <u><b>T1106</b></u><br>Native API                        | <u><b>T1486</b></u><br>Data Encrypted for Impact | <u><b>T1140</b></u><br>Deobfuscate/Decode Files or Information | <u><b>T1204</b></u><br>User execution                  |
| <u><b>T1219</b></u><br>Remote Access Software            | <u><b>T1021</b></u><br>Remote Services           | <u><b>T1083</b></u><br>File and Directory Discovery            |                                                        |

## Indicators of Compromise (IOCs)

| TYPE        | VALUE                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------|
| <b>MD5</b>  | D069812AA63B631897498621DE353519<br>42A5798608F196CE7376CE196F4452FE<br>F365A8BDFD9B39C4F8B9D99613818207 |
| <b>IPV4</b> | 103[.]79[.]76[.]40                                                                                       |

## References

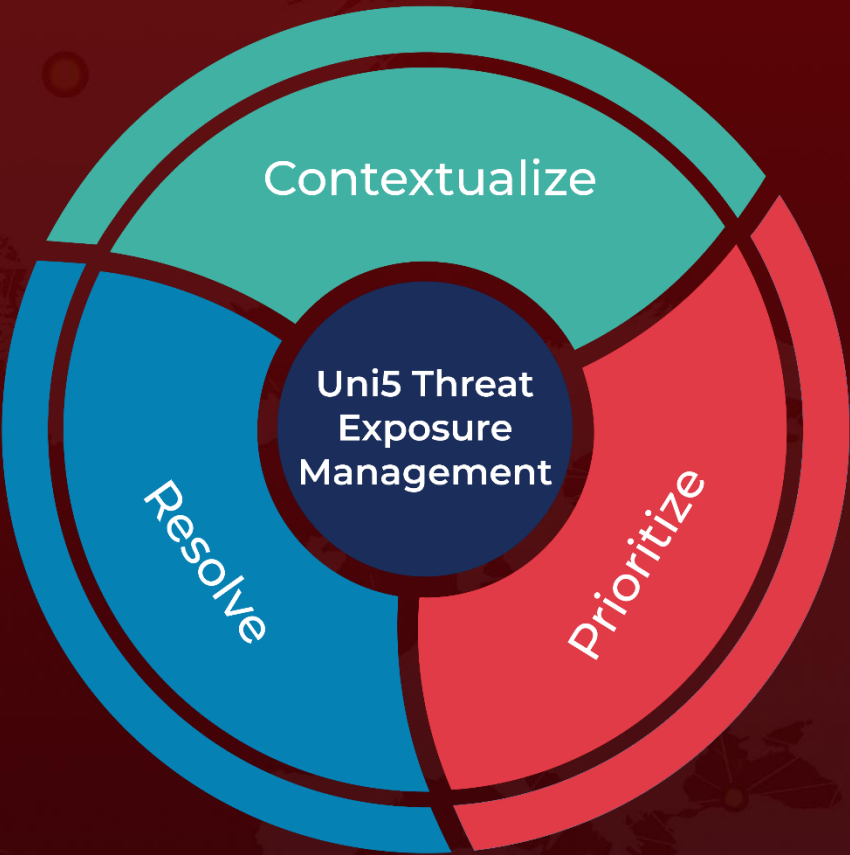
<https://labs.k7computing.com/index.php/pupy-rat-hiding-under-werfaults-cover/>

<https://www.bleepingcomputer.com/news/security/hackers-abuse-windows-error-reporting-tool-to-deploy-malware/>

# What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

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